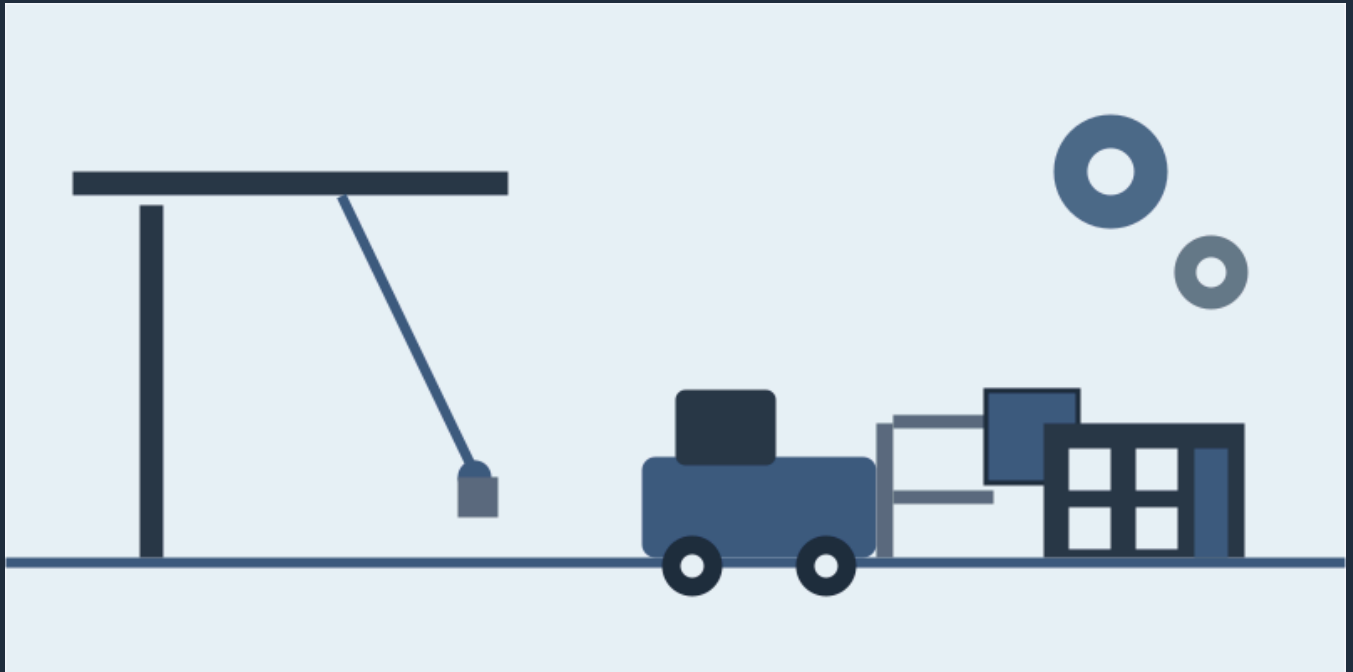




EQUIPIFY

B2B Equipment Rental Platform

Project Proposal



Presented by

CS GROUP 49

SCS2301 / IS2201 — Group Project I

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1. Introduction to Project

The construction and industrial sectors in Sri Lanka rely heavily on specialized equipment, yet purchasing it outright is often not practical for most businesses. Equipment rental is therefore a common necessity. However, no centralized digital platform currently exists in Sri Lanka to connect equipment owners, renters, and skilled operators in a structured way. Businesses are forced to rely on informal networks and phone calls, which is slow, unreliable, and lack transparency.

This project proposes a web-based B2B equipment rental platform that solves these challenges. It will serve as a centralized marketplace where businesses can list, discover, and rent equipment of all types, while also connecting customers with freelance operators, delivery personnel, and maintenance technicians. Key features include rental request management, large order coordination, in-app communication, secure payments, and a rating system to maintain trust across all parties.

2. Project Goals

The goal of this project is to design and develop a fully functional, web-based B2B equipment rental platform that digitizes and streamlines the equipment rental process for businesses in Sri Lanka. By the end of the project, the following outcomes will be accomplished:

- **A centralized rental marketplace** — Equipment owners can list assets, and businesses can discover and rent equipment quickly, replacing informal methods with a transparent digital system.
- **Access to freelance workers** — Customers can connect with equipment operators, delivery personnel, and maintenance technicians all in one place.
- **Large order collaboration** — Equipment owners can collaborate to fulfill orders that a single owner cannot serve alone.
- **Secure online payments** — All financial transactions are handled safely within the platform.

3. Project Scope

The scope of this project clearly defines what the system will include (In-Scope) and what will be excluded (Out-of-Scope) in order to ensure the development remains focused, feasible, and manageable within the project timeline.

This project focuses on designing and developing a web-based B2B Equipment Rental Platform that digitizes the traditional equipment rental process in Sri Lanka. The system aims to connect equipment owners, business customers, and freelance service providers through a single integrated digital platform.

3.1 In-Scope

- B2B equipment rental web platform
- User roles: Customers, Owners, Freelancers, Admin
- Equipment listing, search, and booking system
- Freelance worker integration (operators, delivery, maintenance)
- Online payment system (rentals, deposits, commissions)
- Notification system (booking, payment, status updates)
- Admin dashboard for full system control
- Rental Order Management: The system manages the complete rental process, including equipment requests, approvals, scheduling, and return confirmations.

3.2 Out-Of-Scope

- IoT or GPS tracking hardware integration
- Legal dispute resolution or court handling
- Individual customer rentals
- International / cross-border operations
- Mobile app development

4. Objectives of the Project

The main objective of this project is to develop a web-based B2B Equipment Rental Platform that enables businesses to efficiently rent and manage industrial, construction, event, and media equipment through a centralized system.

Specific objectives are:

- To provide a platform for equipment suppliers to register, manage, and advertise their equipment inventories.
- To enable business customers to search, compare, and rent equipment through an online booking system.
- To implement inventory management features that track equipment availability and rental schedules.
- To support equipment bundles and dependency management for related equipment and accessories.
- To facilitate collaboration among multiple suppliers to fulfill large equipment requests.
- To provide a secure business verification process for suppliers and renters.
- To integrate operator hiring functionality for specialized equipment requiring certified operators.
- To generate analytical reports and dashboards for equipment utilization, revenue tracking, and booking trends.
- To improve efficiency, transparency, and reliability in the equipment rental process for businesses.

5. Project Feasibility

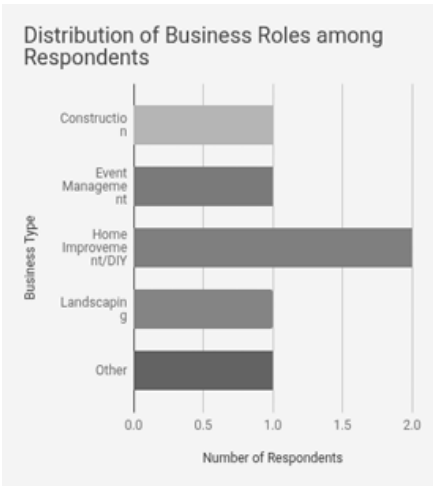
This report evaluates the feasibility of Equipify. This analyses the market feasibility and many factors based on some feedback gathered from actual people across the island through a survey.

You can view the survey structure here: <https://forms.gle/dwxfxGWX3UhuAxp98>

5.1 Operational Feasibility

Operational feasibility analyzes how well our system would suit within a given eco-system of users. Our survey data indicates a diverse range of potential users.

Key Findings



User Diversity

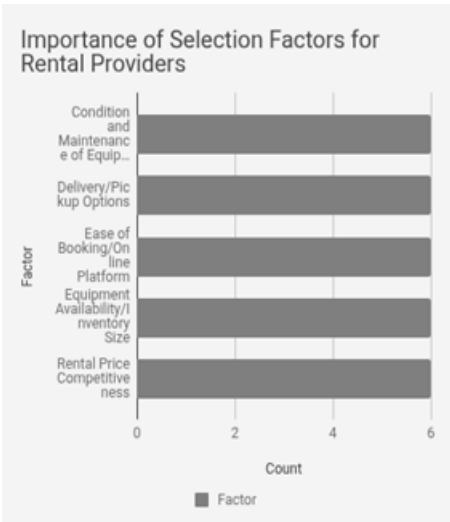
The platform appeals to various sectors, with a notable presence in DIY and construction-related fields.

Critical Success Factors

Respondents identified "Equipment Availability," "Condition and Maintenance," and "Ease of Booking" as the most crucial factors when choosing a rental provider. Equipify’s digital-first approach directly addresses the need for streamlined booking and inventory transparency.

Market Demand

There is a clear demand for a centralized platform that simplifies the rental process and provides reliable equipment.



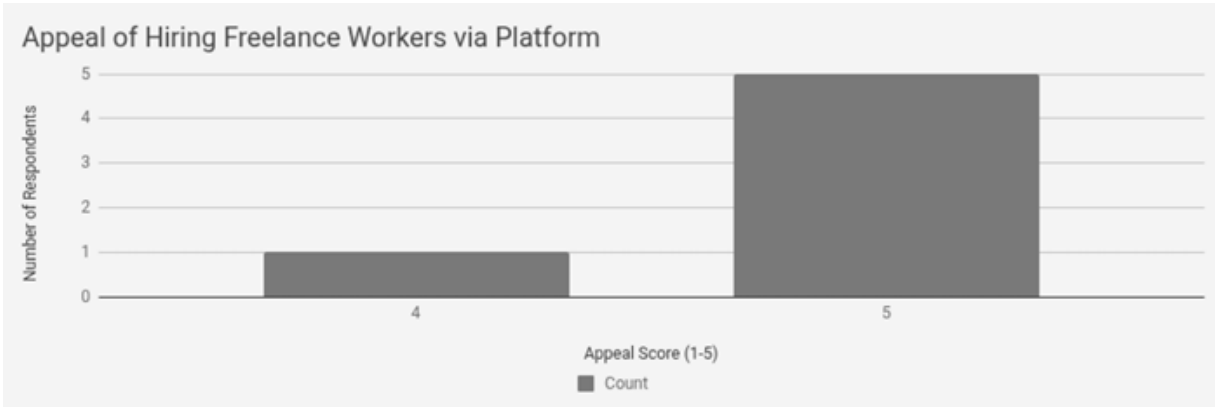
5.2 Technical Feasibility

Technical feasibility focuses on the ability of the organization to build and maintain the platform. Equipify requires a multi-faceted system capable of handling real-time bookings, logistics coordination for delivery, and service integration for maintenance and freelance workers.

Key Findings

Service Integration

A significant majority of respondents expressed high interest in hiring skilled freelance operators directly through the platform. This requires a robust user-matching algorithm and secure communication channels.



Maintenance Support

The data shows that equipment failures frequently cause project delays. The technical requirement to integrate maintenance shops and technicians is not just a "nice-to-have" but a critical solution to a common pain point.

Platform Complexity

While the technical requirements are high (requiring secure payments, geolocation for delivery, and role-based access), they are achievable using modern web and mobile frameworks

5.3 Economic Feasibility

Economic feasibility analyzes the cost-benefit ratio of the project. For Equipify, this involves balancing platform development costs against the potential for transaction fees and service subscriptions.

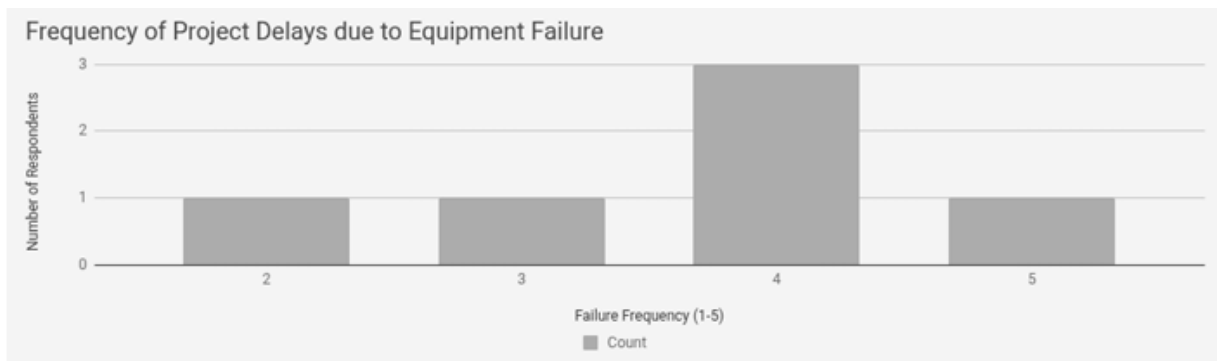
Key Findings

Value Proposition

"Rental Price Competitiveness" is a major factor for users. Equipify can drive prices down by increasing market transparency and competition among providers.

Cost Savings for Users

By providing integrated maintenance and skilled labor, the platform helps users avoid costly project delays. Respondents indicated that the availability of integrated maintenance would significantly increase their likelihood of using the platform.



Revenue Streams

The platform can generate revenue through transaction commissions from rentals and service fees from freelance/maintenance bookings.

5.4 Legal and Regulatory Feasibility

To keep everyone safe and covered, Equipify will use straightforward, standardized rental contracts and clear rules around equipment damage and operator safety.

We'll stick strictly to data privacy laws like GDPR so our users know their information is always secure.

5.5 Scheduling Feasibility

With so many businesses currently facing project delays due to equipment shortages, there is a massive window of opportunity for us to step in right now.

To hit the ground running, we're rolling Equipify out in three agile phases: first, launching our core rental and booking engine; next, bringing delivery and maintenance partners on board; and finally, opening up our freelance worker marketplace and owner analytics.

6. Deliverables of the project

At the completion of the project, the following deliverables will be provided:

- A fully functional B2B Equipment Rental Platform.
- Supplier Management Module.
- Equipment Inventory Management Module.
- Equipment Search and Booking Module.
- Business Verification Module.
- Operator Hiring Module.
- Asset Bundle and Dependency Management Module.
- Analytics and Reporting Dashboard.
- Database Design and Implementation.
- User Documentation and User Manual.
- System Design Documents, including UML diagrams, database schema, and architecture diagrams.
- Project Report and Technical Documentation.
- System Testing and Evaluation Report.
- Source Code Repository and Deployment Instructions.

7. Project Constraints and Assumptions

7.1 Project Constraints

Legal & Compliance Constraints

- When using the system for equipment rental and service tracking, physical or digitally signed documents may be required in certain cases such as disputes, damage claims, or insurance-related issues.
- The system does not directly integrate with government, legal, or regulatory bodies for verification or enforcement purposes. All legal procedures must be handled externally.

Technical Constraints

- The core system will be developed using PHP and MySQL without any frontend or backend frameworks such as Laravel or CodeIgniter.
- Role-based access control (Admin, Customer, Equipment Owner, Freelancers, etc.) will be implemented using custom-built logic without third-party authentication libraries.
- Browser-based notifications and real-time updates are limited by browser compatibility and user permission settings.

Financial Constraints

- The project will be developed using only free-tier services, open-source tools, and locally hosted environments due to the absence of external funding.
- No paid APIs, premium hosting services, or enterprise-level cloud infrastructure will be used.

Organizational Constraints

- The project is strictly limited to the assigned academic group members only. External collaboration or outsourcing is not permitted.
- All project activities including development, documentation, and presentation must follow the academic schedule and evaluation deadlines provided by the institution.

Schedule Constraints

- The entire system must be completed within the academic semester timeline, including analysis, design, implementation, testing, and documentation phases.
- After final submission and evaluation, no further modifications or enhancements will be considered for grading purposes.

Environmental & Operational Constraints

- The system is purely software-based and does not include integration with physical IoT devices or GPS hardware for equipment tracking.
- Real-world operational factors such as traffic delays, weather conditions, and logistics disruptions must be managed manually by users and are not automatically handled by the system.

7.2 Project Assumptions

- It is assumed that all users of the platform (customers, equipment owners, freelance operators, delivery personnel, maintenance workers, and administrators) will have access to a compatible device and a stable internet connection to use the system effectively.
- Users are expected to have basic digital literacy to navigate the platform, including creating accounts, browsing equipment listings, submitting rental requests, and managing bookings.
- It is assumed that equipment owners will actively list their available equipment with accurate and up-to-date information such as availability, pricing, and condition.
- Freelance operators, delivery personnel, and maintenance workers are expected to accept job requests and complete assigned tasks within the agreed time frames.
- It is assumed that all service providers will maintain professional standards and update their task status regularly through the platform.
- The platform is assumed to operate on a commission-based revenue model where a percentage is charged for each successful rental transaction and related service payments.
- It is assumed that administrators and platform managers will already have pre created or verified accounts to manage and monitor the system effectively.
- It is assumed that users will provide accurate personal and business information during registration, which will be used for verification and system operations.

8. Requirements

8.1 Functional Requirements

Customer

- Customers can register and authenticate on the platform.
- Customers can manage their professional profile.
- Customers can browse equipment by category.
- Customers can view equipment details and availability.
- Customers can search and filter equipment listings.
- The system provides smart equipment recommendations based on customer activity.
- Customers can save equipment to a wishlist for later.
- Customers can place a rental request.
- Customers can cancel a pending rental request.
- Customers can request rental extensions and early returns.
- Customers can post large orders.
- Customers can access their full rental history.
- Customers can post job offers for freelance workers.
- Customers can hire a freelance worker.
- Customers can send a delivery request to delivery personnel.
- Customers can accept the delivery personnel's price and confirm the delivery.
- Customers can view commission and platform fee details.
- Customers can view their payment history and invoices.
- Customers can communicate with other parties via in-app messaging.
- Customers can rate other parties.
- The system sends customers both email and in-app notifications for relevant events.

Renting Party

- Renting parties can register and authenticate on the platform.
- Renting parties can manage their professional profile.
- Renting parties can list and manage equipment available for rent.
- Renting parties can set equipment pricing and blackout dates.
- Renting parties can search and filter equipment listings.
- Renting parties can receive and review incoming rental requests.
- Renting parties can confirm equipment return and condition.
- Renting parties can cancel a rental request.
- Renting parties can review and respond to rental extension and early return requests.
- Renting parties can review large orders.
- Renting parties can collaborate with other renting parties to fulfill large orders.
- Renting parties can search and filter maintenance technicians.
- Renting parties can book appointments with maintenance technicians.
- Renting parties can access their full rental history.
- Renting parties can offer in-house delivery for their equipment.
- Renting parties can view their payment history and invoices.
- Renting parties can communicate with other parties via in-app messaging.

- Renting parties can rate other parties.
- The system provides renting parties with an analytics dashboard showing their own rental data.
- The system sends rental status notifications to renting parties.
- The system sends payment notifications to renting parties.

Admin

- Admins can register and authenticate on the platform.
- Admins can manage their profiles.
- Admins can verify credentials submitted by other users.
- Admins can suspend or ban users who violate platform policies.
- Admins can manage platform commission and fee structures.
- Admins can monitor platform activity and user reports.
- The system provides admins with a platform-wide analytics dashboard.
- The system sends admins system and dispute alerts.

Freelance Worker (Equipment Operator)

- Equipment operators can register and authenticate on the platform.
- Equipment operators can manage their professional profile.
- Equipment operators can upload credentials and supporting documents for verification.
- Equipment operators can accept or decline job offers.
- Equipment operators can submit competitive bids on posted job offers.
- Equipment operators can view their job history.
- Equipment operators can showcase a portfolio of work samples on their profile.
- Equipment operators can receive payments through the platform.
- Equipment operators can view their payment history and invoices.
- Equipment operators can communicate with other parties via in-app messaging.
- Equipment operators can rate customers.
- The system sends equipment operators payment notifications.
- The system sends equipment operators job and bid notifications.

Maintenance Technicians

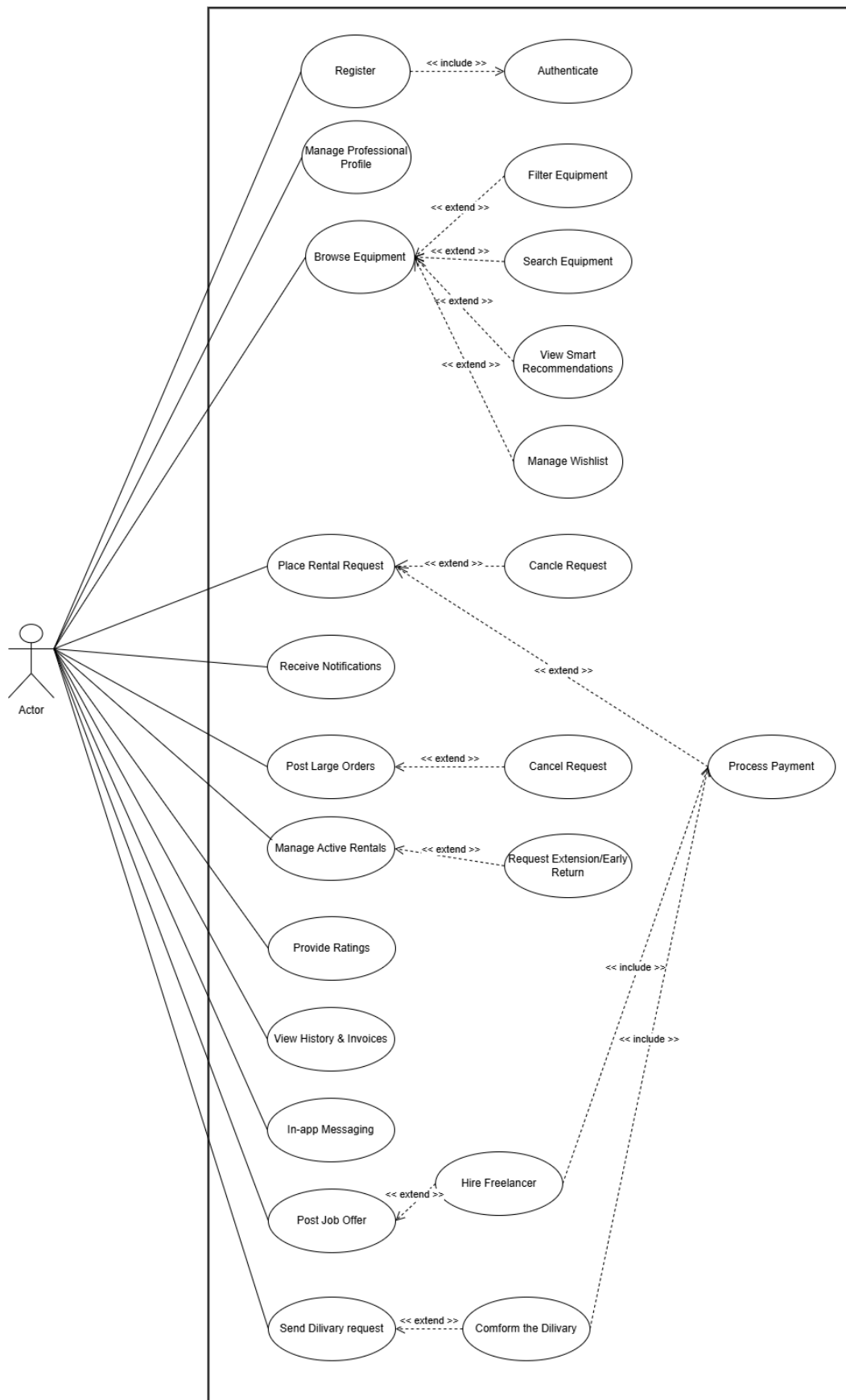
- Maintenance technicians can register and authenticate on the platform.
- Maintenance technicians can manage their professional profile.
- Maintenance technicians can upload credentials and supporting documents for verification.
- Maintenance technicians can list the equipment categories they service and maintain.
- Maintenance technicians can view appointment requests.
- Maintenance technicians can accept or reject appointments.
- Maintenance technicians can view their appointment history.
- Maintenance technicians can showcase a portfolio of work samples on their profile.
- Maintenance technicians can communicate with other parties via in-app messaging.
- The system sends maintenance technicians appointment notifications.

Delivery Personnel

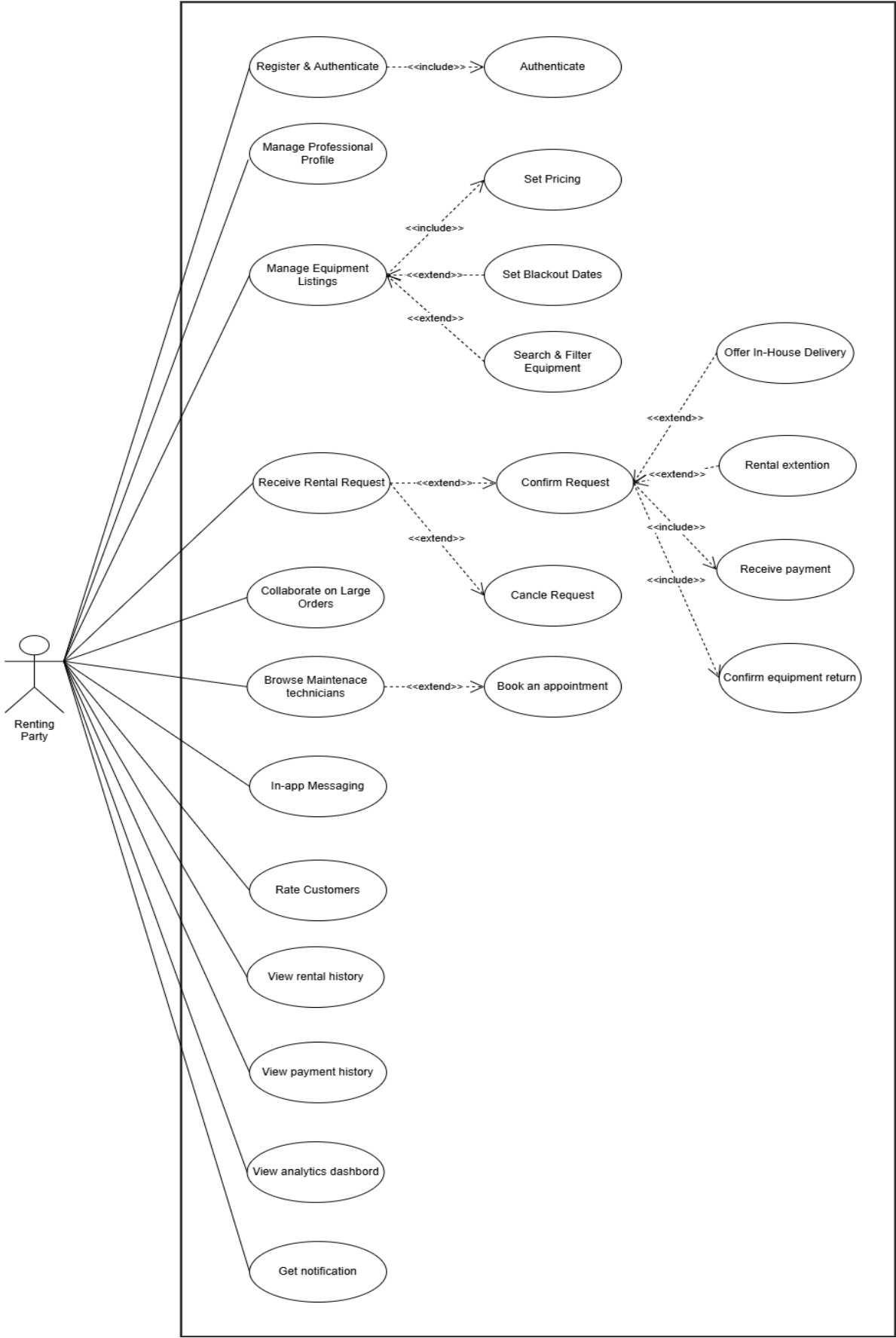
- Delivery personnel can register and authenticate on the platform.
- Delivery personnel can manage their professional profile.
- Delivery personnel can upload credentials and supporting documents for verification.
- Delivery personnel can maintain their vehicle information on the platform.
- Delivery personnel can accept or decline delivery requests.
- Delivery personnel can set a delivery fee for a job.
- Delivery personnel can view their job history.
- Delivery personnel can share their delivery status.
- Delivery personnel can receive payments through the platform.
- Delivery personnel can view their payment history and invoices.
- Delivery personnel can communicate with other parties via in-app messaging.
- Delivery personnel can rate customers.
- The system sends delivery personnel payment notifications.
- The system sends delivery personnel job notifications.

8.1.1 Use Case Diagrams

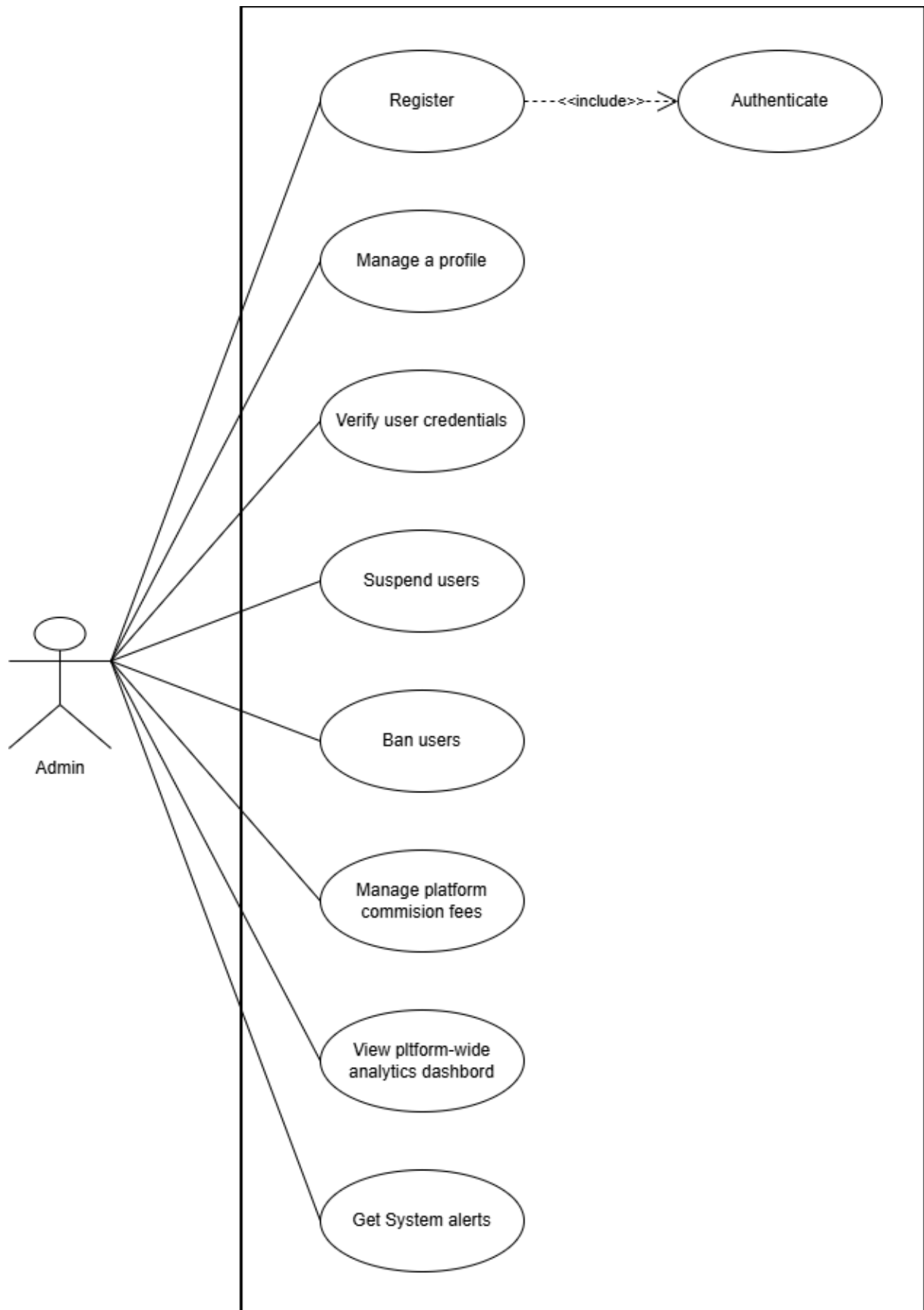
Customer Use Case Diagram



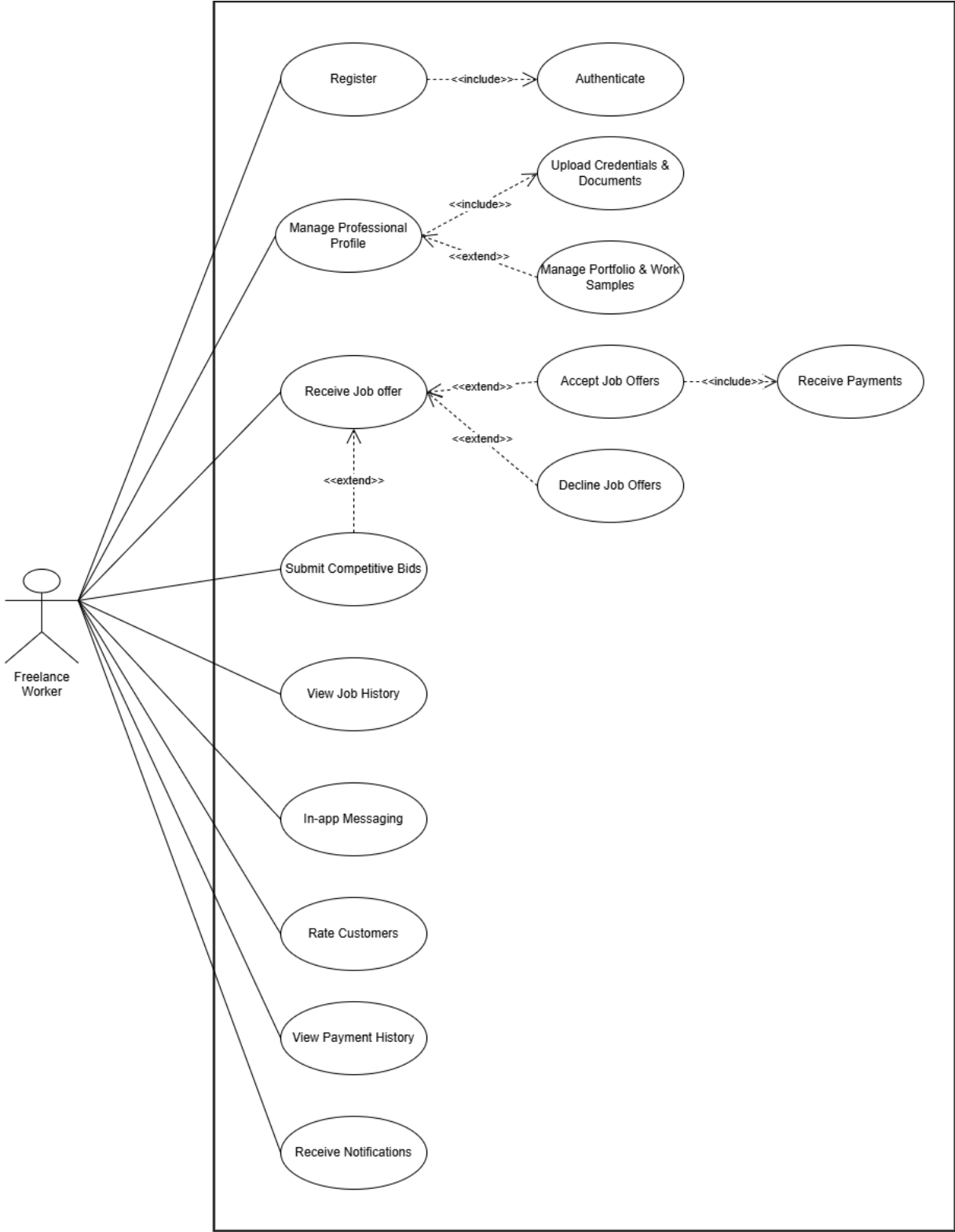
Renting Party Use Case Diagram



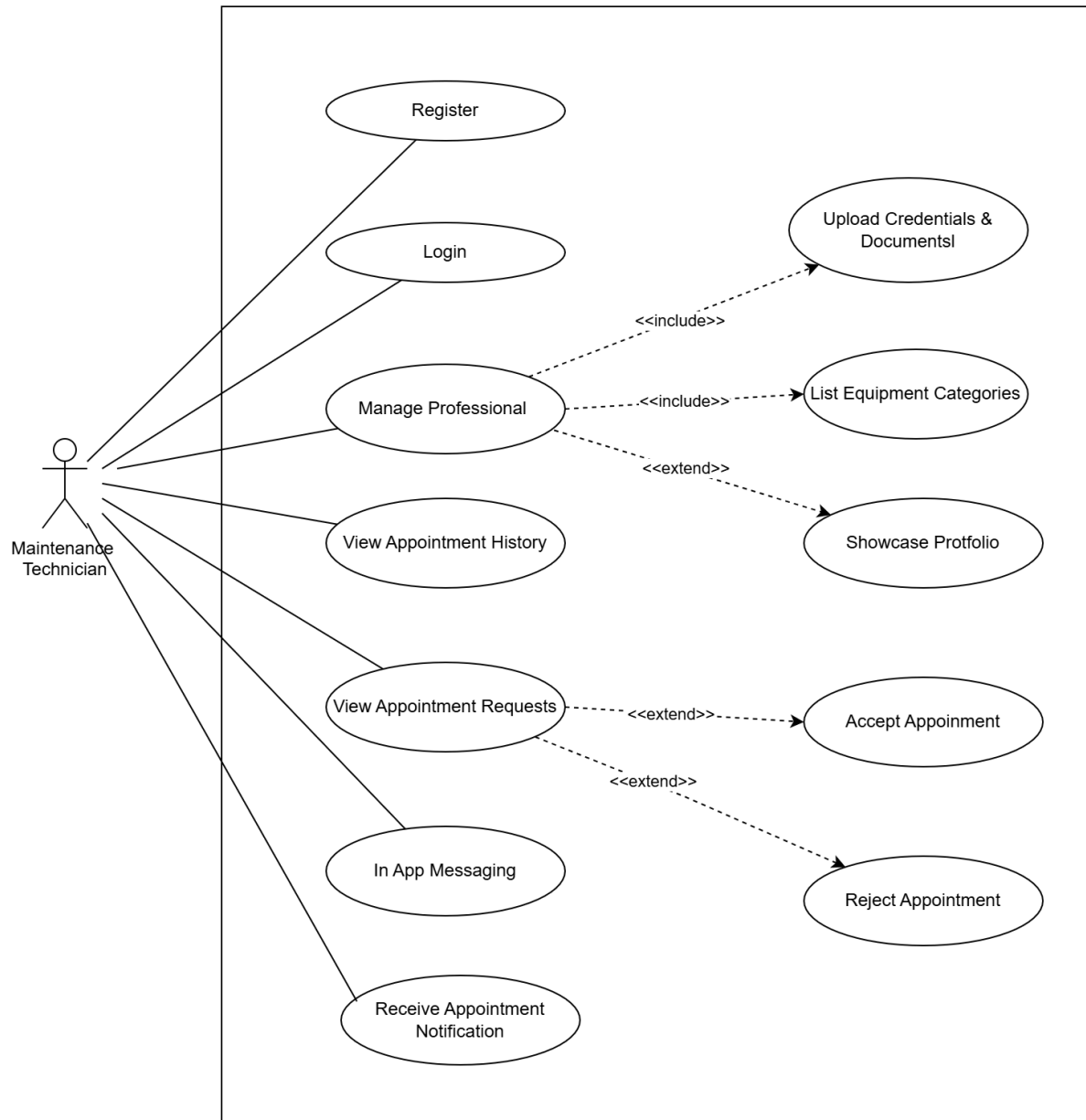
Admin Use Case Diagram



Freelancer Worker Use Case Diagram



Maintenance Technician Use Case Diagram



Delivery Personnel Use Case Diagram



8.1.2 Use Case Description

Shared Use Cases

Use Case ID	01
Title	Register
Primary Actor	All Actors
Description	A new user creates an account on the platform.
Pre Conditions	The user has not previously registered.
Main Flow	1. User navigates to the registration page. 2. User enters required details (name, email, password, role). 3. User submits the form. 4. System validates the details and creates the account.
Alternate Flows	• Email already registered: system prompts the user to log in instead. • Invalid/missing field: system highlights the error and requests correction.
Post Condition	A new account is created and the user is redirected to login.

Use Case ID	02
Title	Login
Primary Actor	All Actors
Description	A registered user logs into the platform.
Pre Conditions	The user has a registered account.
Main Flow	1. User navigates to the login page. 2. User enters email and password. 3. System validates credentials and grants access. 4. System redirects the user to their dashboard.
Alternate Flows	• Invalid credentials: system displays an error. • Forgot password: system provides a reset password option.
Post Condition	User is logged into the system.

Use Case ID	03
Title	Manage Profile
Primary Actor	All Actors
Description	A logged-in user views and updates their profile information.
Pre Conditions	User is authenticated.
Main Flow	1. User navigates to the profile page. 2. User edits required fields.

	3. User saves the changes. 4. System validates and saves the updated profile.
Alternate Flows	Invalid data: system highlights the error and requests correction.
Post Condition	Profile is updated.

Use Case ID	04
Title	Upload Credentials & Documents
Primary Actor	Freelance Worker (Operator), Delivery Personnel, Maintenance Technician
Description	A worker uploads professional credentials and documents for admin verification.
Pre Conditions	User is authenticated and managing their profile.
Main Flow	- User navigates to the credentials section. - User uploads required documents (certifications, ID, licenses). - User submits the documents. - System saves documents and flags them for admin review.
Alternate Flows	Unsupported file format: system displays an error and requests a valid format.
Post Condition	Documents are uploaded and pending admin verification.

Use Case ID	05
Title	In-App Messaging
Primary Actor	All Actors (except Admin, Payment Gateway, Notification Engine)
Description	A user communicates with other parties through the platform's messaging system.
Pre Conditions	User is authenticated and has an active or pending transaction with another party.
Main Flow	- User navigates to the messaging section. - User selects an existing conversation or starts a new one. - User types and sends a message. - System delivers the message and notifies the recipient.
Alternate Flows	Recipient is offline: message is stored and delivered upon their next login.
Post Condition	Message is delivered to the recipient.

Use Case ID	06
Title	Receive Notifications
Primary Actor	All Actors (except Payment Gateway)

Description	User receives real-time in-app and email notifications for relevant platform events.
Pre Conditions	User is registered on the platform.
Main Flow	1. A platform event occurs (rental update, payment, job offer, appointment, etc.). 2. Notification engine generates a notification. 3. System delivers it via in-app alert and email. 4. User views the notification.
Alternate Flows	• User is offline: notification is stored and displayed on next login.
Post Condition	User is informed of the relevant event.

Use Case ID	07
Title	Receive Payments
Primary Actor	Renting Party, Freelance Worker (Operator), Delivery Personnel
Description	A user receives payment for a completed service through the platform's payment gateway.
Pre Conditions	A rental, job, or delivery has been confirmed and completed.
Main Flow	1. System initiates payment processing upon confirmation. 2. Payment gateway processes the transaction. 3. Platform deducts the applicable commission fee. 4. Remaining amount is credited to the user's account. 5. System sends a payment notification.
Alternate Flows	• Payment fails: system notifies both parties and cancels the associated action.
Post Condition	Payment is processed and credited to the user's account.

Use Case ID	08
Title	View Payment History
Primary Actor	Customer, Renting Party, Freelance Worker (Operator), Delivery Personnel
Description	A user views their past payment records and invoices.
Pre Conditions	User is authenticated and has at least one completed transaction.
Main Flow	1. User navigates to the payment history page. 2. System displays all payment records with dates and amounts. 3. User selects a record to view details or download the invoice.
Alternate Flows	• No records exist: system displays a message indicating no payment history found.
Post Condition	User views payment records and can download invoices.

Use Case ID	09
Title	View Job History
Primary Actor	Freelance Worker (Operator), Delivery Personnel
Description	A worker views their past completed and cancelled jobs.
Pre Conditions	User is authenticated and has at least one past job.
Main Flow	1. User navigates to the job history page. 2. System displays all past jobs with details and status. 3. User selects a job to view full details.
Alternate Flows	• No history exists: system displays a message indicating no records found.
Post Condition	User views their job history.

Use Case ID	10
Title	Rate Customers
Primary Actor	Renting Party, Freelance Worker (Operator), Delivery Personnel
Description	A user submits a rating for a customer after a completed transaction.
Pre Conditions	User has completed a rental or job with the customer.
Main Flow	1. User navigates to the completed transaction. 2. User selects the option to leave a rating. 3. User provides a star rating and optional written review. 4. User submits the rating. 5. System saves and publishes the rating on the customer's profile.
Alternate Flows	• Rating already submitted: system notifies the user that a rating already exists.
Post Condition	Rating is saved and visible on the customer's profile.

Customer Use Cases

Use Case ID	01
Title	Browse Equipment
Primary Actor	Customer
Description	Customer browses available equipment listings by category.
Pre Conditions	Customer is authenticated.
Main Flow	1. Customer navigates to the equipment browsing page. 2. System displays equipment grouped by category. 3. Customer selects a category to view listings.
Alternate Flows	• No equipment in selected category: system displays no results message.
Post Condition	Customer views a list of available equipment.

Use Case ID	02
Title	Search & Filter Equipment
Primary Actor	Customer, Renting Party
Description	User searches for equipment using keywords and applies filters to narrow results.
Pre Conditions	User is on the equipment browsing page.
Main Flow	1. User enters a keyword or selects filter options (price, location, availability). 2. System applies the search/filter criteria. 3. System displays matching results.
Alternate Flows	• No matching results: system displays a no results message.
Post Condition	System displays equipment listings matching the criteria.

Use Case ID	03
Title	View Smart Recommendations
Primary Actor	Customer
Description	System suggests relevant equipment based on the customer's activity.
Pre Conditions	Customer is authenticated.
Main Flow	1. Customer navigates to the equipment page. 2. System analyses activity and rental history. 3. System displays recommended equipment.
Alternate Flows	• No prior activity: system displays popular listings instead.
Post Condition	Customer views recommended equipment.

Use Case ID	04
Title	Manage Wishlist
Primary Actor	Customer
Description	Customer saves or removes equipment listings from a wishlist.
Pre Conditions	Customer is authenticated and viewing an equipment listing.
Main Flow	1. Customer clicks the save to wishlist button on a listing. 2. System adds the listing to the wishlist. 3. Customer can view all saved items on the wishlist page.
Alternate Flows	• Item already in wishlist: system removes it instead.
Post Condition	Equipment is added to or removed from the wishlist.

Use Case ID	05
Title	Place Rental Request
Primary Actor	Customer
Description	Customer submits a request to rent a specific piece of equipment.
Pre Conditions	Customer is authenticated and viewing an available equipment listing.
Main Flow	1. Customer selects a listing and clicks Request Rental. 2. Customer enters rental details (dates, delivery preference). 3. Customer reviews summary and confirms. 4. System submits the request and triggers Process Payment. 5. System notifies the renting party.
Alternate Flows	• Payment fails: system cancels the request and notifies the customer.
Post Condition	Rental request is submitted and renting party is notified.

Use Case ID	06
Title	Cancel Rental Request
Primary Actor	Customer, Renting Party
Description	User cancels a pending rental request.
Pre Conditions	User has an active pending rental request.
Main Flow	1. User navigates to active rental requests. 2. User selects the request to cancel. 3. User confirms the cancellation. 4. System cancels the request and notifies the other party.
Alternate Flows	• Request already confirmed: system applies the cancellation policy and may initiate a refund.
Post Condition	Rental request is cancelled and both parties are notified.

Use Case ID	07
Title	Manage Active Rentals
Primary Actor	Customer
Description	Customer views and manages currently active rental agreements.
Pre Conditions	Customer has at least one active rental.
Main Flow	1. Customer navigates to the active rentals page. 2. System displays active rentals with details. 3. Customer selects a rental to view or take action.
Alternate Flows	• No active rentals: system displays a no active rentals message.
Post Condition	Customer views or updates active rental status.

Use Case ID	08
Title	Request Extension / Early Return
Primary Actor	Customer
Description	Customer requests to extend the rental period or return equipment early.
Pre Conditions	Customer has an active rental agreement.
Main Flow	1. Customer selects Request Extension or Early Return on an active rental. 2. Customer enters the new date and a reason. 3. Customer confirms and submits. 4. System notifies the renting party.
Alternate Flows	• Renting party rejects: system notifies customer and original agreement remains.
Post Condition	Request is submitted and renting party is notified.

Use Case ID	09
Title	Post Large Order
Primary Actor	Customer
Description	Customer posts a large equipment order that may require multiple renting parties.
Pre Conditions	Customer is authenticated.
Main Flow	1. Customer navigates to the large order posting page. 2. Customer enters order details (equipment types, quantities, dates, location). 3. Customer submits the order. 4. System publishes the order and notifies relevant renting parties.
Alternate Flows	• Missing required field: system highlights the error and requests completion.
Post Condition	Large order is posted and visible to renting parties.

Use Case ID	10
Title	Post Job Offer
Primary Actor	Customer
Description	Customer posts a job offer for freelance operators to apply for.
Pre Conditions	Customer is authenticated and has an active or upcoming rental.
Main Flow	1. Customer navigates to the post job offer page. 2. Customer enters job details (equipment type, location, dates, requirements). 3. Customer submits the offer. 4. System publishes the offer and notifies relevant freelancers.
Alternate Flows	• Missing required field: system highlights the error and requests completion.
Post Condition	Job offer is posted and visible to freelancers.

Use Case ID	11
Title	Hire Freelancer
Primary Actor	Customer
Description	Customer selects and hires a freelancer from received bids.
Pre Conditions	Customer has a posted job offer with at least one bid.
Main Flow	1. Customer reviews received bids on the job offer. 2. Customer selects a preferred freelancer. 3. Customer confirms the hiring decision. 4. System triggers Process Payment. 5. System notifies the hired freelancer and closes remaining bids.
Alternate Flows	• Payment fails: hiring is cancelled and customer is notified.
Post Condition	Freelancer is hired and job is confirmed.

Use Case ID	12
Title	Send Delivery Request
Primary Actor	Customer
Description	Customer requests delivery of rented equipment through delivery personnel.
Pre Conditions	Customer has a confirmed rental requiring delivery.
Main Flow	1. Customer navigates to the delivery request page. 2. Customer enters pickup and drop-off location. 3. System estimates the delivery fee based on distance. 4. Customer submits the request. 5. System notifies available delivery personnel.

Alternate Flows	No delivery personnel available: system notifies the customer.
Post Condition	Delivery request is sent to available delivery personnel.

Use Case ID	13
Title	Confirm Delivery
Primary Actor	Customer
Description	Customer accepts a delivery fee quote and confirms the delivery job.
Pre Conditions	Customer has an active delivery request with at least one quote received.
Main Flow	- Customer receives a delivery fee quote. - Customer reviews and accepts the quote. - System triggers Process Payment. - System notifies the delivery person that the job is confirmed.
Alternate Flows	- Customer rejects the quote: request stays open for other delivery personnel. - Payment fails: confirmation is cancelled and customer is notified.
Post Condition	Delivery is confirmed and delivery person is assigned.

Use Case ID	14
Title	Process Payment
Primary Actor	Customer, Payment Gateway
Description	System processes a payment on behalf of the customer through the payment gateway.
Pre Conditions	Customer has confirmed a rental, hired a freelancer, or confirmed a delivery.
Main Flow	- System initiates a payment request to the payment gateway. - Customer confirms payment details. - Payment gateway processes the transaction. - System records the transaction and notifies relevant parties.
Alternate Flows	- Payment declined: system notifies the customer and cancels the associated action. - Network error: system retries before displaying an error.
Post Condition	Payment is processed and transaction record is created.

Use Case ID	15
Title	View Rental History
Primary Actor	Customer, Renting Party
Description	User views past rental transactions.
Pre Conditions	User is authenticated and has at least one completed rental.

Main Flow	1. User navigates to the rental history page. 2. System displays all past rentals with details. 3. User selects a rental to view full details.
Alternate Flows	• No history: system displays no records found message.
Post Condition	User views their rental history.

Use Case ID	16
Title	Provide Ratings
Primary Actor	Customer
Description	Customer submits a rating for a renting party or freelancer after a completed transaction.
Pre Conditions	Customer has completed a rental or hired a freelancer.
Main Flow	1. Customer navigates to the completed transaction. 2. Customer selects the rate option. 3. Customer provides a star rating and optional review. 4. Customer submits the rating. 5. System saves and publishes the rating.
Alternate Flows	• Rating already submitted: system notifies the customer.
Post Condition	Rating is saved and visible on the rated party's profile.

Renting Party Use Cases

Use Case ID	01
Title	Manage Equipment Listings
Primary Actor	Renting Party
Description	Renting party adds, edits, or removes equipment listings.
Pre Conditions	Renting party is authenticated.
Main Flow	1. Renting party navigates to equipment listings. 2. Renting party adds a new listing or edits an existing one. 3. Renting party enters equipment details and sets pricing. 4. Renting party submits the listing. 5. System saves and publishes the listing.
Alternate Flows	• Missing required field: system highlights error and requests completion. • Remove listing: system unpublishes it after confirmation.
Post Condition	Equipment listing is created, updated, or removed.

Use Case ID	02
Title	Set Pricing
Primary Actor	Renting Party
Description	Renting party sets the rental price for a piece of equipment.
Pre Conditions	Renting party is managing an equipment listing.
Main Flow	1. Renting party enters rental rates (daily, weekly, or monthly). 2. Renting party confirms pricing. 3. System saves pricing to the listing.
Alternate Flows	• Invalid price (zero or negative): system displays error and requests correction.
Post Condition	Pricing is saved to the equipment listing.

Use Case ID	03
Title	Set Blackout Dates
Primary Actor	Renting Party
Description	Renting party marks dates when equipment is unavailable for rental.
Pre Conditions	Renting party has an active equipment listing.
Main Flow	1. Renting party selects Set Blackout Dates on a listing. 2. Renting party selects dates to mark as unavailable. 3. Renting party confirms selection. 4. System saves blackout dates and updates the availability calendar.

Alternate Flows	Dates overlap with confirmed rental: system alerts the renting party and blocks the action.
Post Condition	Selected dates are marked as unavailable.

Use Case ID	04
Title	Receive & Confirm Rental Request
Primary Actor	Renting Party
Description	Renting party reviews and confirms an incoming rental request.
Pre Conditions	Renting party has received a rental request from a customer.
Main Flow	1. System notifies renting party of a new request. 2. Renting party reviews the request details. 3. Renting party confirms the request. 4. System triggers Receive Payment. 5. System notifies the customer of the confirmation. 6. Upon rental completion, system triggers Confirm Equipment Return.
Alternate Flows	- Payment fails: confirmation is cancelled and both parties are notified. - No response within allowed time: system auto-cancels and notifies customer.
Post Condition	Rental is confirmed and customer is notified.

Use Case ID	05
Title	Review Rental Extension / Early Return
Primary Actor	Renting Party
Description	Renting party reviews and responds to a customer's extension or early return request.
Pre Conditions	Renting party has an active rental with a pending extension or early return request.
Main Flow	1. System notifies renting party of the request. 2. Renting party reviews the request details. 3. Renting party approves or rejects. 4. System updates the rental and notifies the customer.
Alternate Flows	Rejected: system notifies customer and original agreement remains.
Post Condition	Rental agreement is updated accordingly.

Use Case ID	06
Title	Confirm Equipment Return
Primary Actor	Renting Party
Description	Renting party confirms equipment has been returned and records its condition.

Pre Conditions	Rental period has ended and equipment has been returned.
Main Flow	1. Renting party selects Confirm Return on the active rental. 2. Renting party records the equipment condition. 3. Renting party confirms the return. 4. System closes the rental and notifies the customer.
Alternate Flows	• Equipment returned damaged: renting party flags the issue and system initiates a dispute.
Post Condition	Return is confirmed and rental is closed.

Use Case ID	07
Title	Offer In-House Delivery
Primary Actor	Renting Party
Description	Renting party offers their own delivery service for equipment to customers.
Pre Conditions	Renting party has a confirmed rental requiring delivery.
Main Flow	1. Renting party enables the in-house delivery option. 2. Renting party sets the delivery fee. 3. System presents the option to the customer. 4. Customer accepts or declines.
Alternate Flows	• Customer declines: customer may use platform delivery personnel instead.
Post Condition	In-house delivery option is presented to the customer.

Use Case ID	08
Title	Collaborate on Large Orders
Primary Actor	Renting Party
Description	Renting party partners with other renting parties to collectively fulfill a large order.
Pre Conditions	A customer has posted a large order.
Main Flow	1. Renting party views available large orders. 2. Renting party expresses interest in fulfilling part of the order. 3. System matches renting parties to collectively fulfill the order. 4. All parties confirm participation. 5. System notifies the customer.
Alternate Flows	• Insufficient renting parties: system notifies the customer.
Post Condition	Large order is collectively assigned to collaborating renting parties.

Use Case ID	09
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Title	Browse & Book Maintenance Technician
Primary Actor	Renting Party
Description	Renting party searches for and books a maintenance technician.
Pre Conditions	Renting party is authenticated.
Main Flow	1. Renting party navigates to the maintenance technicians page. 2. Renting party searches or filters by equipment category or location. 3. System displays matching technician profiles. 4. Renting party selects a technician and enters appointment details. 5. Renting party submits the booking request. 6. System notifies the technician.
Alternate Flows	• No technicians available: system displays no results message. • Technician rejects: system notifies renting party and suggests alternatives.
Post Condition	Appointment request is submitted and technician is notified.

Use Case ID	10
Title	View Analytics Dashboard
Primary Actor	Renting Party
Description	Renting party views their rental performance data and insights.
Pre Conditions	Renting party is authenticated and has rental activity.
Main Flow	1. Renting party navigates to the analytics dashboard. 2. System displays rental statistics (total rentals, revenue, utilization, ratings). 3. Renting party applies date filters if needed.
Alternate Flows	• No activity: system displays no data available message.
Post Condition	Renting party views their analytics data.

Admin Use Cases

Use Case ID	01
Title	Verify User Credentials
Primary Actor	Admin
Description	Admin reviews and verifies credentials submitted by users.
Pre Conditions	A user has uploaded credentials requiring verification.
Main Flow	1. Admin navigates to the credential verification page. 2. System displays pending verification requests. 3. Admin reviews submitted documents. 4. Admin approves or rejects the credentials. 5. System updates the user's verification status and notifies them.
Alternate Flows	• Rejected: system notifies the user and requests resubmission.
Post Condition	User's credentials are marked as verified or rejected.

Use Case ID	02
Title	Suspend Users
Primary Actor	Admin
Description	Admin temporarily suspends a user account due to a policy violation.
Pre Conditions	Admin has identified a user requiring suspension.
Main Flow	1. Admin navigates to user management and selects a user. 2. Admin selects Suspend. 3. Admin enters a reason and confirms. 4. System suspends the account and notifies the user.
Alternate Flows	• User has active rentals: system alerts admin before proceeding.
Post Condition	User account is suspended.

Use Case ID	03
Title	Ban Users
Primary Actor	Admin
Description	Admin permanently bans a user account from the platform.
Pre Conditions	Admin has identified a user requiring a permanent ban.
Main Flow	1. Admin navigates to user management and selects a user. 2. Admin selects Ban. 3. Admin enters a reason and confirms. 4. System permanently revokes access and notifies the user.
Alternate Flows	• Active rentals or pending payments: system alerts admin to resolve them first.

Post Condition	User account is permanently banned.
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Use Case ID	04
Title	Manage Platform Commission Fees
Primary Actor	Admin
Description	Admin configures and updates the platform's commission and fee structures.
Pre Conditions	Admin is authenticated.
Main Flow	1. Admin navigates to the commission and fee management page. 2. Admin reviews current fee structures. 3. Admin updates rates or amounts. 4. Admin confirms and saves. 5. System applies updated fees to all future transactions.
Alternate Flows	• Invalid values: system highlights error and requests correction.
Post Condition	Updated fees are saved and applied to future transactions.

Use Case ID	05
Title	Monitor Activity & Reports
Primary Actor	Admin
Description	Admin monitors platform-wide activity, user reports, and analytics.
Pre Conditions	Admin is authenticated.
Main Flow	1. Admin navigates to the monitoring or analytics dashboard. 2. System displays platform-wide metrics and flagged reports. 3. Admin reviews data and applies filters as needed. 4. Admin takes appropriate action on flagged items.
Alternate Flows	• No data for selected filters: system displays no data found message.
Post Condition	Admin reviews activity and acts on reports as needed.

Use Case ID	06
Title	Resolve Disputes
Primary Actor	Admin
Description	Admin responds to user queries and resolves disputes between parties.
Pre Conditions	A dispute or query has been raised by a user.
Main Flow	1. System sends admin a dispute alert. 2. Admin reviews the dispute details. 3. Admin contacts the involved parties via messaging. 4. Admin makes a resolution decision. 5. System applies the resolution and notifies both parties.

Alternate Flows	Escalation required: admin escalates to a higher authority.
Post Condition	Dispute is resolved and parties are notified.

Freelance Worker (Operator) Use Cases

Use Case ID	01
Title	Manage Portfolio & Work Samples
Primary Actor	Freelance Worker (Operator), Maintenance Technician
Description	Worker adds or updates work samples on their profile to showcase experience.
Pre Conditions	User is authenticated and managing their profile.
Main Flow	- User navigates to the portfolio section. - User uploads work samples (images, descriptions). - User saves the portfolio. - System publishes the portfolio on their public profile.
Alternate Flows	Unsupported file format: system displays error and requests a valid format.
Post Condition	Portfolio is updated and visible on the public profile.

Use Case ID	02
Title	Receive Job Offer
Primary Actor	Freelance Worker (Operator)
Description	Freelancer receives a job offer posted by a customer.
Pre Conditions	A customer has posted a job offer matching the freelancer's skills.
Main Flow	- System notifies the freelancer of a new job offer. - Freelancer navigates to the job offers page. - Freelancer reviews job details. - Freelancer proceeds to Accept, Decline, or Submit a Bid.
Alternate Flows	Expired offer: system displays offer no longer available message.
Post Condition	Freelancer has reviewed and acted on the job offer.

Use Case ID	03
Title	Accept Job Offer
Primary Actor	Freelance Worker (Operator)
Description	Freelancer accepts a job offer, triggering payment processing.
Pre Conditions	Freelancer has reviewed a job offer and chosen to accept.
Main Flow	Freelancer selects Accept.

	2. Freelancer confirms acceptance. 3. System triggers Receive Payments. 4. System notifies the customer.
Alternate Flows	Payment fails: acceptance is cancelled and both parties are notified.
Post Condition	Job is confirmed and customer is notified.

Use Case ID	04
Title	Decline Job Offer
Primary Actor	Freelance Worker (Operator)
Description	Freelancer declines a job offer they are unable or unwilling to take.
Pre Conditions	Freelancer has reviewed a job offer.
Main Flow	1. Freelancer selects Decline. 2. Freelancer enters an optional reason. 3. System records the decline and notifies the customer.
Alternate Flows	Already accepted by another worker: system informs the freelancer.
Post Condition	Job offer is declined and customer is notified.

Use Case ID	05
Title	Submit Competitive Bid
Primary Actor	Freelance Worker (Operator)
Description	Freelancer submits a bid on a posted job offer.
Pre Conditions	A job offer is available and open for bids.
Main Flow	1. Freelancer views an open job offer. 2. Freelancer selects Submit Bid. 3. Freelancer enters proposed rate and a short pitch. 4. Freelancer submits the bid. 5. System records the bid and notifies the customer.
Alternate Flows	Bidding period closed: system displays bids no longer accepted message.
Post Condition	Bid is submitted and customer is notified.

Delivery Personnel Use Cases

Use Case ID	01
Title	Maintain Vehicle Information
Primary Actor	Delivery Personnel
Description	Delivery person maintains vehicle details on their profile.
Pre Conditions	Delivery person is authenticated and managing their profile.
Main Flow	1. Delivery person navigates to the vehicle information section. 2. Delivery person enters or updates vehicle details (type, model, plate number, capacity). 3. Delivery person saves the information. 4. System updates the profile.
Alternate Flows	• Invalid vehicle data: system highlights error and requests correction.
Post Condition	Vehicle information is updated on the profile.

Use Case ID	02
Title	Manage Deliveries
Primary Actor	Delivery Personnel
Description	Delivery person accepts or declines delivery requests, sets fees, and updates delivery status.
Pre Conditions	Delivery person has received at least one delivery request.
Main Flow	1. Delivery person navigates to the deliveries page. 2. System displays incoming delivery requests. 3. Delivery person reviews request details (pickup, drop-off, distance). 4. Delivery person sets their delivery fee. 5. Delivery person accepts or declines the request. 6. Upon accepting, delivery person updates status as the job progresses.
Alternate Flows	• Declined: system notifies customer and keeps request open for others. • Unable to complete: delivery person updates status and notifies customer.
Post Condition	Delivery request is accepted or declined and status is updated.

Maintenance Technician Use Cases

Use Case ID	01
Title	List Equipment Categories
Primary Actor	Maintenance Technician
Description	Technician specifies equipment categories they are qualified to service.
Pre Conditions	Technician is managing their professional profile.
Main Flow	1. Technician navigates to the equipment categories section. 2. Technician selects the categories they service. 3. Technician saves the selection. 4. System updates the profile with the selected categories.
Alternate Flows	• No category selected: system prompts technician to select at least one.
Post Condition	Equipment categories are saved and displayed on the technician's profile.

Use Case ID	02
Title	View Appointment Requests
Primary Actor	Maintenance Technician
Description	Technician views and responds to incoming appointment requests.
Pre Conditions	Technician has received at least one appointment request.
Main Flow	1. System notifies technician of a new appointment request. 2. Technician navigates to appointment requests. 3. Technician reviews request details (equipment type, date, time, location). 4. Technician proceeds to Accept or Reject the appointment.
Alternate Flows	• No response within allowed time: system auto-cancels and notifies renting party.
Post Condition	Technician has reviewed and acted on the appointment request.

Use Case ID	03
Title	Accept Appointment
Primary Actor	Maintenance Technician
Description	Technician accepts an appointment request from a renting party.
Pre Conditions	Technician has a pending appointment request.
Main Flow	1. Technician selects Accept. 2. Technician confirms acceptance. 3. System confirms the appointment and notifies the renting party. 4. Appointment is added to the technician's schedule.
Alternate Flows	• Time slot conflict: system alerts technician and asks them to choose another slot.

Post Condition	Appointment is confirmed and renting party is notified.
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Use Case ID	04
Title	Reject Appointment
Primary Actor	Maintenance Technician
Description	Technician rejects an appointment request they are unable to fulfill.
Pre Conditions	Technician has a pending appointment request.
Main Flow	1. Technician selects Reject. 2. Technician enters an optional reason. 3. System records the rejection and notifies the renting party.
Alternate Flows	• Payment already made: system initiates a refund before closing the request.
Post Condition	Appointment is rejected and renting party is notified.

Use Case ID	05
Title	View Appointment History
Primary Actor	Maintenance Technician
Description	Technician views their past completed and cancelled appointments.
Pre Conditions	Technician has at least one past appointment.
Main Flow	1. Technician navigates to the appointment history page. 2. System displays all past appointments with details and status. 3. Technician selects an appointment to view full details.
Alternate Flows	• No history: system displays no records found message.
Post Condition	Technician views their appointment history.

Use Case ID	06
Title	Receive Appointment Notification
Primary Actor	Maintenance Technician
Description	Technician receives real-time notifications for new appointment requests and updates.
Pre Conditions	Technician is registered on the platform.
Main Flow	1. A platform event occurs (new appointment request, cancellation, etc.). 2. Notification engine generates a notification. 3. System delivers notification via in-app alert and email. 4. Technician views the notification.
Alternate Flows	• Technician is offline: notification is stored and shown on next login.
Post Condition	Technician is informed of the appointment event.

8.2) Quality attributes Requirements

1. Security and Privacy

- The platform handles sensitive corporate data, financial transactions, and personal verification documents, making security a critical requirement.

2. Performance and Efficiency

- The system must remain responsive during equipment searches, freelancer bidding, and delivery tracking.

3. Availability and Reliability

- Unplanned downtime could result in financial losses for businesses that depend on the platform daily.

4. Usability and Accessibility

- The platform serves users with varying levels of technical proficiency, from corporate managers to on-site gig workers. The interface will be fully responsive across desktops, tablets, and mobile devices.

5. Scalability

- The system must handle growth in users, equipment listings, and concurrent activity without performance degradation.

6. Maintainability and Extensibility

- The codebase must be easy to manage, update, and extend throughout the project lifecycle.

9. Technologies to be used

9.1 PHP

Chosen PHP as the server-side language considering its scalability and robustness. It supports the OOP structure and we'll be utilizing it to the maximum to make the code maintainable and easy to understand as a whole.

9.2 MySQL

MySQL is the perfect fit for this project because its relational structure fits hand-in-glove with our object-oriented code. By using an ORM, we can seamlessly map our PHP classes directly to MySQL tables, keeping our data and application logic perfectly in sync. Plus, MySQL makes it incredibly easy to scale as we grow; through simple indexing, replication, and sharding, it can effortlessly handle heavy traffic and complex data relationships without breaking a sweat or compromising on data integrity.

9.3 HTML, CSS & JavaScript

Using HTML, CSS, and JavaScript is a no-brainer because they are the universal languages of the web. Every modern browser like Chrome, Safari, and Edge is built to run them flawlessly right out of the box without needing any extra plugins. By sticking to these native standards, we can guarantee a smooth, consistent, and fast experience for our users, whether they are on a flagship desktop or a budget smartphone.

9.4 WebSockets

Using WebSockets would let us leverage stateful connections with the backend to handle real-time communication and related features like handling notifications and chat features. This is different from the traditional request-response model and lets us utilize the power of persistent connections.

9.6 OpenStreetMap API

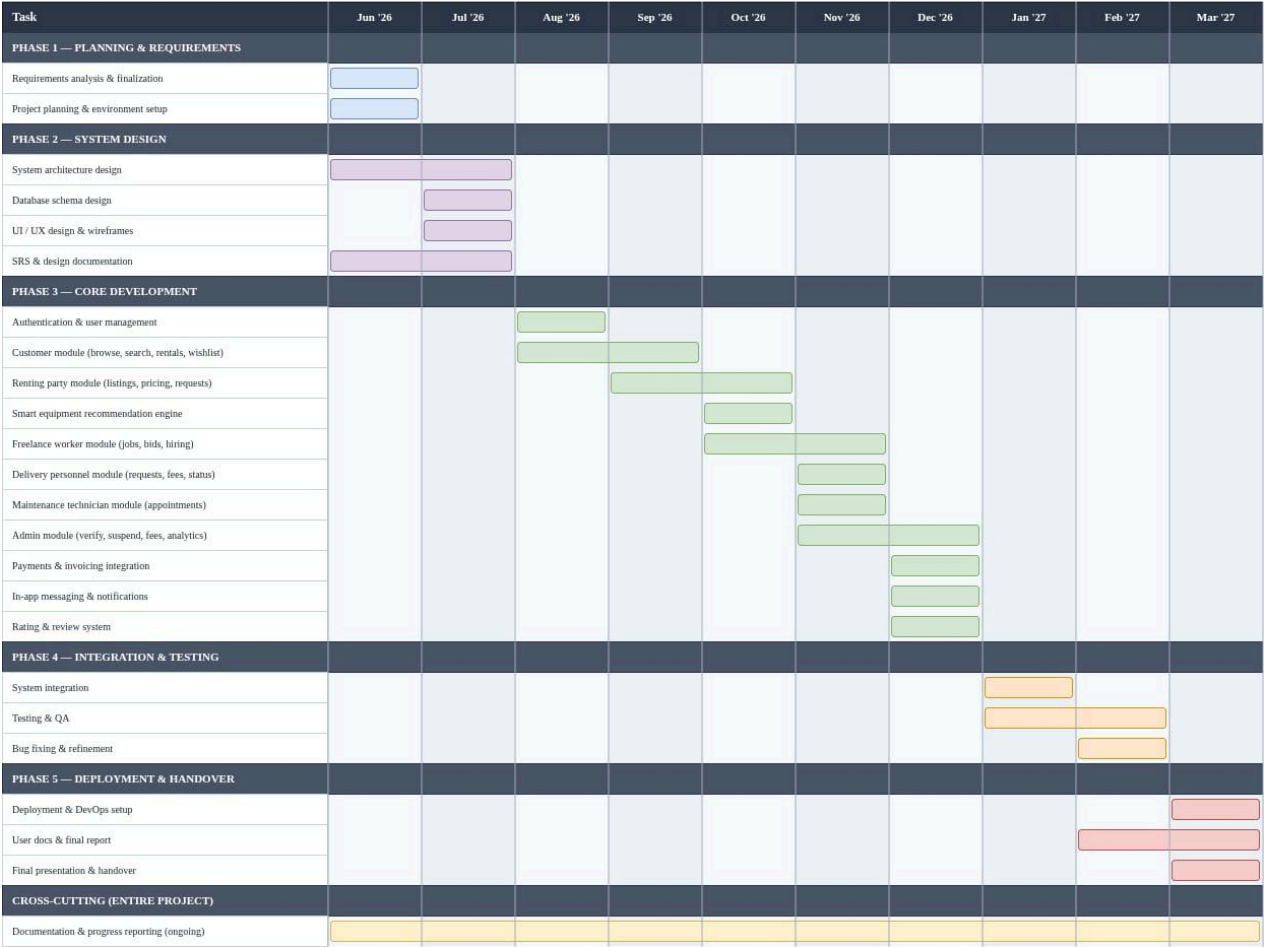
We'll be using the OpenStreetMap API to calculate the real-time distance between two coordinates in real time. This is the most cost effective option available having zero-cost being an open source platform.

9.7 PayHere Payment Gateway

Using PayHere as the payment gateway for the platform to handle all the transactions. There is an official sandbox and an SDK available for PHP and convenient in using out of the box within a testing environment.

10. Project timeline

B2B Hardware Equipment Rental Platform | Project Gantt Chart (Jun 2026 – Mar 2027)



Legend (colour = project phase):

Planning & Requirements

System Design

Core Development

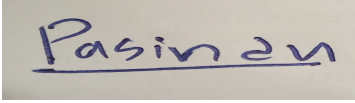
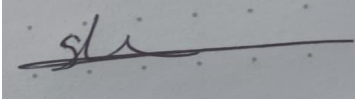
Integration & Testing

Deployment & Handover

Cross-cutting (ongoing)

11. Declaration

We as members of the project titled **Equipify**, Certify that we will carry out this project according to the guidelines provided by the coordinators and supervisors of the course as well as we will not incorporate, without acknowledgment, any material previously submitted for a degree or diploma in any university. To the best of our knowledge and brief, the project work will not contain any material previously published or written by another person or ourselves except where due reference is made in the text of appropriate places.

Index Number	Name of the student	Signature
K.W.P.N. Vidmantha	24002119	
T. S. I. Peiris	24001546	
K.T.D. Rangajeewa	24001716	
S. Dishanthan	24000485	